Group 5 - Brazilian E-Commerce EDA

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Project Overview

This project analyses Brazil's e-commerce ecosystem using the Olist public dataset, which covers orders placed on the Olist marketplace between 2016 and 2018. Olist connects small and medium sellers to major Brazilian marketplaces through a single contract, making it a representative sample of Brazil's digital economy. The analysis is framed around SDG 8, examining how logistics performance, regional inequality, customer retention and payment behaviour relate to sustainable economic growth and decent work for sellers.

Research Questions

- Does delivery timing affect customer satisfaction?
- Are late deliveries concentrated in specific regions?
- Which product categories retain customers best and is e-commerce growing?
- How do Brazilians prefer to pay and what does it reveal about financial inclusion?

Methodology

1. Data Loading 9 Olist tables loaded and merged into analysis-ready dataframes using inner/left joins on order_id and customer_id.
2. Data Overview Null counts, data types and row counts documented for all tables. Targeted cleaning applied per insight**,** only delivered orders used for delivery-time analyses, comment nulls in reviews left as-is.
3. Exploratory Analysis Five chart types across four insights: line chart, bar chart, horizontal bar chart, box plot and scatter plot. Each visualization followed by plain-English interpretation.
4. Export Cleaned master dataframe (95,824 rows, 16 columns) exported as cleaned_dataset.csv covering delivered orders only.

Key Findings

Insight 1 : Order Trends
-Order volume tripled from around 300/month (Oct 2016) to 7,400/month (Nov 2017).
-Brazilians shop most on Mondays and during afternoons (12-18h).
-Black Friday 2017 produced the single largest monthly spike in the dataset.
-SDG 8.1: sustained digital economic growth confirmed across the period.

Insight 2 : Delivery Delay vs Review Score
-Early deliveries average 4.29 stars: orders 10+ days late average only 1.69.
-6.5% of all orders were late, but they generate a disproportionate share of 1-star reviews.
-A 2.6-point drop on a 5-point scale purely from delivery timing.
-SDG 8.2: logistics quality directly limits seller productivity and income.

Insight 3 : Geographic Late Delivery

-AL (20.6%) and MA (16.7%) have the highest late delivery rates in Brazil.
-Northeastern states dominate the top 15 list : Brazil's poorest region.
-Remote states (RO, AM) have low late rates because estimates are padded conservatively. •
-SDG 8.1: regional logistics inequality mirrors and reinforces economic inequality.

Insight 4 : Repeat Purchase by Category
-Only 3% of customers placed more than one order, the platform is in acquisition mode.
-home_appliances (17.8%) and fashion_bags_accessories (13%) lead repeat rates.
-books_technical (2.3%) and computers (3%) have the lowest repeat rates.
-SDG 8.3: low retention means fragile seller revenue and limited small business growth.

Conclusion

Brazil's e-commerce market shows strong growth momentum aligned with SDG 8, but two structural barriers limit its potential as an engine of inclusive economic growth. Logistics inequality between regions means sellers and customers in the Northeast face systematically worse delivery performance, directly suppressing satisfaction scores and seller income. Low customer retention (3%) means platform revenue for sellers is dependent on constant new-customer inflow rather than sustainable loyalty. Improving last-mile delivery in underserved regions and building conditions for repeat purchasing are the two highest-impact interventions for making Brazilian digital commerce a genuine engine of decent work.

SDG 8 Alignment

Finding	SDG 8 Target
Late deliveries hurt seller income	8.2 Economic productivity
Northeast worst delivery rates	8.1 Regional growth
3% repeat rate, low retention	8.3 Small business growth
E-commerce tripling 2016-2018	8.1 Sustained growth

Limitations

• The dataset covers 2016-2018 only. Brazil's e-commerce market grew sharply after 2020 and current patterns may differ significantly.
• Repeat purchase analysis is limited by the dataset window. A customer who bought in late 2018 had no time to return before data ends.
• The analysis identifies associations, not causal effects. Delivery delay correlates with low scores but other unmeasured factors may contribute.
• The dataset does not include seller-side data such as seller revenue or churn, which would allow stronger SDG 8 claims about employment.

Data Source

- Olist. Brazilian E-Commerce Public Dataset. Kaggle, 2018. kaggle.com/datasets/olistbr/brazilian-ecommerce
- The Global Goals. *Goal 8: Decent Work and Economic Growth.* United Nations, 2015. https://globalgoals.org/goals/8-decent-work-and-economic-growth/